KAIZEN MATHS

Lesson starter

Differential Equations: First and Second Order

Second-Order Constant Coefficient

Name _____

S1

Solve each second-order differential equation with distinct real auxiliary roots.

$y'' - y' - 6y = 0$

S2

Solve each second-order differential equation with repeated auxiliary roots.

$y'' + 2y' + y = 0$

S3

Solve each second-order differential equation with complex auxiliary roots.

$y'' - 2y' + 10y = 0$

S4

Solve each non-homogeneous second-order differential equation.

$y'' + 9y = 27$

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Exit ticket

Differential Equations: First and Second Order

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Name _____

E1

Solve each second-order differential equation with distinct real auxiliary roots.

$y'' - 5y' + 6y = 0$

E2

Solve each second-order differential equation with repeated auxiliary roots.

$y'' + 6y' + 9y = 0$

E3

Solve each second-order differential equation with complex auxiliary roots.

$y'' + 16y = 0$

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Show your working

Teacher answers & working

Differential Equations: First and Second Order · Second-Order Constant Coefficient

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S1

Solve each second-order differential equation with distinct real auxiliary roots.

$y'' - y' - 6y = 0$

Answer:

$y = Ae^{-2x} + Be^{3x}$

Working:

Form the auxiliary equation by replacing y'' with m^2, y' with m, and y with 1.

$m^2 - m - 6 = 0$

Factorise the auxiliary equation.

$(m + 2)(m - 3) = 0$

$m = -2$ or $m = 3$

For distinct real roots, use $y = Ae^{m_1x} + Be^{m_2x}$.

$y = Ae^{-2x} + Be^{3x}$

S2

Solve each second-order differential equation with repeated auxiliary roots.

$y'' + 2y' + y = 0$

Answer:

$y = (A + Bx)e^{-x}$

Working:

Form the auxiliary equation.

$m^2 + 2m + 1 = 0$

Factorise the auxiliary equation.

$(m + 1)^2 = 0$

$m = -1$, repeated root

For a repeated root, use $y = (A + Bx)e^{mx}$.

$y = (A + Bx)e^{-x}$

Teacher answers & working

Differential Equations: First and Second Order · Second-Order Constant Coefficient

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S3

Solve each second-order differential equation with complex auxiliary roots.

$y'' - 2y' + 10y = 0$

Answer:

$y = e^{x}(A \cos 3x + B \sin 3x)$

Working:

Form the auxiliary equation.

$m^2 - 2m + 10 = 0$

Solve the auxiliary equation.

$m = 1 \pm 3i$

For roots $\alpha \pm \beta i$, use $y = e^{\alpha x}(A \cos \beta x + B \sin \beta x)$.

$y = e^{x}(A \cos 3x + B \sin 3x)$

S4

Solve each non-homogeneous second-order differential equation.

$y'' + 9y = 27$

Answer:

$y = A \cos 3x + B \sin 3x + 3$

Working:

First solve the complementary equation $y'' + 9y = 0$.

Auxiliary equation: $m^2 + 9 = 0$

$m = \pm 3i$

Complementary function: $y_c = A \cos 3x + B \sin 3x$

Because the right side is constant, try a constant particular solution $y_p = k$.

$0 + 9k = 27$

$k = 3$

Add the complementary function and particular integral.

$y = A \cos 3x + B \sin 3x + 3$

Teacher answers & working

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E1

Solve each second-order differential equation with distinct real auxiliary roots.

$y'' - 5y' + 6y = 0$

Answer:

$y = Ae^{2x} + Be^{3x}$

Working:

Form the auxiliary equation by replacing y'' with m^2, y' with m, and y with 1.

$m^2 - 5m + 6 = 0$

Factorise the auxiliary equation.

$(m - 2)(m - 3) = 0$

$m = 2$ or $m = 3$

For distinct real roots, use $y = Ae^{m_1x} + Be^{m_2x}$.

$y = Ae^{2x} + Be^{3x}$

E2

Solve each second-order differential equation with repeated auxiliary roots.

$y'' + 6y' + 9y = 0$

Answer:

$y = (A + Bx)e^{-3x}$

Working:

Form the auxiliary equation.

$m^2 + 6m + 9 = 0$

Factorise the auxiliary equation.

$(m + 3)^2 = 0$

m = -3, repeated root

For a repeated root, use $y = (A + Bx)e^{mx}$.

$y = (A + Bx)e^{-3x}$

Teacher answers & working

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E3

Solve each second-order differential equation with complex auxiliary roots.

$y'' + 16y = 0$

Answer:

$y = A\cos 4x + B\sin 4x$

Working:

Form the auxiliary equation.

$m^2 + 16 = 0$

Solve the auxiliary equation.

$m = 0 \pm 4i$

For roots $\alpha \pm \beta i$, use $y = e^{\alpha x}(A\cos \beta x + B\sin \beta x)$.

$y = A\cos 4x + B\sin 4x$